# AICHU TAN

**Data Scientist | Machine Learning | AI Engineer**

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## PROFESSIONAL SUMMARY

Data Scientist with hands-on experience developing machine learning, statistical modeling, and AI applications across public health, agriculture, and urban analytics. Experienced building end-to-end data pipelines, predictive models, computer vision systems, and LLM-powered applications using Python, SQL, TensorFlow, LangChain, AWS, and Streamlit. Passionate about transforming complex data into actionable business insights through scalable, production-ready solutions.

## TECHNICAL SKILLS

**Programming:** Python, SQL, R, JavaScript, C++
**Data Analysis & Visualization:** Pandas, NumPy, Data Cleaning, Data Wrangling, Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering, Tableau, Matplotlib, Streamlit, ArcGIS
**SQL & Databases:** Advanced SQL (Joins, CTEs, Window Functions), ETL, Data Modeling
**Machine Learning:** Scikit-learn, XGBoost, Random Forest, Classification, Regression, Clustering, K-Prototypes, Cross-Validation, Hyperparameter Tuning
**Deep Learning & AI:** TensorFlow, LSTM, YOLOv8, Computer Vision, LangChain, Retrieval-Augmented Generation (RAG)
**Time Series Forecasting:** ARIMA, SARIMA, SARIMAX, Exponential Smoothing, Forecasting
**Cloud & MLOps:** AWS (EC2, S3, RDS), MLflow, Git, Hugging Face Spaces, Supabase, Render

## PROFESSIONAL EXPERIENCE

### Data Scientist
*Sep 2025 - Present*

*SDSU DiMoLab*

Portfolio: Brief | Demo | GitHub

- Designed end-to-end SQL and Python ETL pipelines to ingest and standardize 27+ years of longitudinal meteorological, entomological, and population data for outbreak modeling.
- Developed automated data-quality validation rules to flag anomalies and filter reporting noise before downstream modeling.
- Diagnosed instability in daily-frequency case counts and re-aggregated data into 3-day and 5-day windows, stabilizing forecasts and achieving **$R^2$** up to **0.89**, enabling earlier, more targeted public health response.
- Deployed an interactive forecasting application using Streamlit, Supabase, and Render Cloud, enabling users to explore forecasts, compare model performance, and visualize prediction results.
- Co-authoring a research manuscript on outbreak forecasting for submission to a peer-reviewed journal.

### AI Engineer
*Feb 2025 - Dec 2025*

*SDSU Metabolism of Cities Living Lab*

Portfolio: Brief | Demo | GitHub

- Designed and deployed an interactive AI platform on Hugging Face Spaces using YOLOv8, LangChain RAG, and Streamlit, enabling real-time plant disease diagnosis and AI-powered treatment recommendations.
- Analyzed model performance across 35,000 plant disease images using confusion matrix analysis and improved annotation quality for 13,500+ images, increasing detection accuracy from **70%** to **89.4% mAP@50.**
- Built an LLM-powered retrieval system grounded in agricultural research literature to minimize hallucinations and deliver reliable, evidence-based recommendations while reducing API costs.
- Co-authored research accepted at the 4th International One Health Conference (Rome, 2025).

## SELECTED DATA SCIENCE PROJECTS

### Predicting Restaurant Ratings Using Yelp Metadata

*Python • SQL • XGBoost • Random Forest • MLflow*

- Developed an end-to-end machine learning classification pipeline using 36,261 Yelp restaurant records to predict high- versus low-rated businesses based on operational attributes rather than review text.
- Engineered 61 predictive features from business metadata, including operating hours, service offerings, and accessibility; achieved **73.3% accuracy** and **0.79 ROC-AUC** using XGBoost, with experiment tracking in MLflow.
- Found via K-Prototypes clustering to identify customer behavior patterns, revealing that accessibility features were stronger predictors of restaurant ratings than price tier, providing actionable business insights.

### Online Ticket Reservation System

*MySQL • Python • AWS RDS*

- Designed a normalized relational database (ERD → schema) with trigger-based validation to guarantee transactional data integrity across bookings and payments.
- Deployed the database on AWS RDS with role-based access control, enabling secure and scalable data management.

- Built a Python/Datapane dashboard surfacing **$70K+** in revenue trends and customer behavior patterns for business decision-making.

**Mapping for Change - Urban Inequality Analysis**

*Spatial Analysis • Time Series • Data Visualization*

- Integrated five public datasets (ACS, 311, PITC, and Tree Equity Score) into an interactive ArcGIS dashboard to analyze housing burden, homelessness, and environmental equity across urban neighborhoods.
- Performed exploratory data analysis and spatial statistical analysis to identify significant geographic clusters linking housing cost burden with reduced tree equity, supporting targeted policy recommendations.

## OTHER EXPERIENCE

**Business Owner -** *Wholesale Clothing* *Apr 2009 - Oct 2016*

- Increased annual sales **50%** through customer segmentation, market analysis, and demand forecasting.
- Optimized inventory planning and supply chain operations using sales forecasting and purchasing analytics.
- Managed international sourcing and import/export operations across Taiwan, China, and Indonesia.

**Process Engineer -** *ProMOS Technology* *Jul 2004 - Feb 2009*

- Applied statistical process control (SPC), root cause analysis, and data analysis to improve semiconductor manufacturing yield.
- Reduced manufacturing defects through statistical modeling, process optimization, and cross-functional problem solving.

## RESEARCH & PUBLICATIONS

**Climate-Driven Lag Modeling for Dengue Control Policy in Taiwan**

Co-author • *Manuscript in preparation (2026)*

**AI for Farmers: A YOLO and Language Model-Based Plant Disease Detection and Advisory System**

Aichu Tan, Domenico Vito, Gabriel Fernandez

*Submitted to One Health Advances (Springer Nature); under peer review (2026)*

**Influence of Low Temperature Barrier Nitride Film on Transistor Characteristics for Advanced DRAM**

Dewi Hamza *(former name)*, C. Wang, P. Lin, C.-K. Kao, C. Kuo, An Ku

*24th International VLSI Multilevel Interconnection Conference (VMIC 2007)* | 2007 | pp. 451–455

**Racemization and Hydrolysis of (S)-Naproxen 2,2,2-Trifluoroethyl Ester in Non-Polar Solvents**

Dewi Hamza (former name), Han-Yuan Lin, Eddy Lay, Wen-Yen Wen, Yu-Chi Cheng, Shau-Wei Tsai

*Journal of Physical Organic Chemistry*, 2004

## EDUCATION

**M.S. in Big Data Analytics**, San Diego State University - **GPA 4.0** *Aug 2024 - Dec 2025*

**Google Data Analytics Professional Certificate** *Dec 2022*

**B.S. in Chemical Engineering**, National Cheng Kung University (Taiwan) *Sept 1999 - June 2003*